UNIT 8: ADVANCED INTEGRATION TECHNIQUES

In this section, we will introduce two more commonly employed integration techniques: Integration by Parts and Integration by Partial Fractions. We will also revisit L'Hôpital's Rule, and briefly discuss more geometric applications of integration, including arc length and surface area.

So far (in Calc AB), you should be familiar with u-substitution (reverse chain rule) to compute integrals.
As a quick refresher:

$$\int x sec^2(x^2)\, dx = \tfrac{1}{2} tan\,(x^2) + C$$

$$\int tan(x)\, dx = \int \frac{sin\,(x)}{cos\,(x)}\, dx = -\, ln\,|cos\,(x)| + C$$

$$\int x tan\,(x^2)\, dx = -\tfrac{1}{2} ln\,|cos\,(x^2)\,| + C$$

But there are of course some integrals that cannot be solved through u-sub. For example:

$$\int x^2\, ln\,(x)\, dx$$

Think about it. What would you substitute for x? Or x^2?

But there is another integration technique that allows us to solve this integral.

<u>Integration by Parts</u>

We know by the product rule, for a function y = uv, such that u and v are both functions of x,

$$\frac{dy}{dx} = u\,(\frac{dv}{dx}) + v\,(\frac{du}{dx})$$

Multiplying both sides by dx yields ...

$$dy = u\,(dv) + v\,(du)$$

And now if we were to integrate both sides of the equation (without considering the addition of a constant) ...

$$\int dy = \int u\,(dv) + v\,(du)$$

$$y = \int u\,(dv) + \int v\,(du)$$

And since we defined y = uv ...

$\int u\,(dv) = uv - \int v\,(du)$

This is the Integration by Parts Formula

Confused? Don't worry; let's work through a few problems first:

Example 1: $\int x^2 \ln(x)\,dx$

First, let's figure out which function will be u " and which function will be " dv ".
For now, let's say that u = $\ln(x)$ and dv = x^2 — <u>we will elaborate more on this particular decision later.</u>

u = $\ln(x)$ v = $\frac{1}{3}x^3$

du = $\frac{1}{x}$ **dv** = $x^2 dx$

So according to the formula $\int u\,(dv) = uv - \int v\,(du)$

$$\int x^2 \ln(x)\,dx = \tfrac{1}{3}x^3 \ln(x) - \tfrac{1}{3}\int x^3 (\tfrac{1}{x})\,dx$$

Still following? Now, solving the integral on the right-hand side should be pretty straightforward:

$$\int x^2 \ln(x)\,dx = \tfrac{1}{3}x^3 \ln(x) - \tfrac{1}{3}\int x^3 (\tfrac{1}{x})\,dx$$

$$= x^3 \ln(x) - \tfrac{1}{3}\int x^2\,dx$$

$$= x^3 \ln(x) - \tfrac{1}{3}(\tfrac{1}{3})x^3 + C$$

So, our final answer is just $x^3 \ln(x) - \frac{1}{9}x^3$ + C (remember the constant of integration!)

And we're done!! We can check, however by differentiating our answer:

$$\frac{d}{dx}\left(\tfrac{1}{3}x^3 \ln(x) - \tfrac{1}{9}x^3 + C\right) = x^2 \ln(x) + \tfrac{1}{3}x^2 - \tfrac{1}{2}x^2 = x^2 \ln(x)$$

<u>Note that by differentiating the antiderivative, we arrived back at the function that we originally started with.</u>

Now what if we had instead assigned v = $\ln(x)$ and u = x^2 ?

Nothing, actually — it'll just be a much harder integral to solve:

u = x^2 v = $xln(x) - x$ *

du = $2x\,dx$ **dv** = $ln\,(x)$

*This integral itself requires integration by parts -- we will actually use this example later on.

So, $\int x^2\,ln\,(x)\,dx = (x^3 ln\,(x) - x^3) - 2\int x^2 ln\,(x) - x^2\,dx$

Evidently, this integral would require us to use integration by parts multiple times. In short, it becomes more complicated, and it would be a lot easier to compute had we differentiated $ln\,(x)$ instead.

For the sake of simplicity, there is a general rule you want to follow when determining which function to differentiate (or which function is assigned *u*):

Logarithmic (since the derivative of $\log_b(x) = \frac{1}{x\,ln(b)}$, or essentially $\frac{1}{x}$ multiplied by a constant)
Inverse trigonometric
Polynomial
Exponential
Trigonometric

Keep this mnemonic in mind as we continue to solve more examples.

Example 2: $\int xcos(x)dx$

Let's choose to differentiate *x* and integrate *cos(x)*

u = x v = $cos(x)$

du = 1 **dv** = $sin\,(x)$

Now we integrate:

$$\int xcos(x)dx = xsin(x) - \int sin(x)\,dx = xsin(x) + cos(x) + C$$

To check, let's differentiate our answer:

$$\frac{d}{dx}\,(\,xsin(x) + cos(x) + C) = sin(x) + xcos(x) - sin(x) = xcos(x)\,,$$

This is the original function we integrated.

Exercise 8.1

Try these problems by yourself first before reading the solutions below:

1. $\int x^3 e^x \, dx$ **Hint:** *In some cases, you may have to integrate more than once*

2. $\int cos(x) e^x \, dx$

3. $\int ln\,(x)\, dx$ **Hint:** $\int 1 \, dx = x + C$

Solutions:

1. $x^3 e^x - 3x^2 e^x + 6x e^x - 6e^x + C$ (final answer)

Steps:

$u = x^x$ $v = e^x$

$du = 3x^2$ $dv = e^x$

$\int x^3 e^x \, dx = x^3 e^x - 3\int x^2 e^x \, dx$

$u = x^2$ $v = e^x$

$du = 2x$ $dv = e^x$

$\int x^3 e^x \, dx = x^3 e^x - 3\int x^2 e^x \, dx = x^3 e^x - 3\,(x^2 e^x - 2\int x e^x \, dx)$

$u = x$ $v = e^x$

$du = 1$ $dv = e^x$

$\int x^3 e^x \, dx = x^3 e^x - 3\int x^2 e^x \, dx = x^3 e^x - 3\,(x^2 e^x - 2\int x e^x \, dx) = x^3 e^x - 3x^2 e^x + 6(x e^x - \int e^x \, dx)$

$= x^3 e^x - 3x^2 e^x + 6x e^x - 6e^x$

Thus, $\int x^3 e^x \, dx = x^3 e^x - 3x^2 e^x + 6x e^x - 6e^x + C$

2. $\frac{1}{2}e^x\,(sin(x) + cos(x)) + C$ (final answer)

Steps:

We can start this the same way we would solve any integral:

u = e^x v = $sin\,(x)$

du = e^x dv = $cos(x)$

$$\int cos(x)e^x\,dx = sin(x)\,e^x - \int sin(x)e^x\,dx$$

Let's use integration by parts again:

u = e^x v = $-cos\,(x)$

du = e^x dv = $sin(x)$

$$\int cos(x)e^x\,dx = sin(x)\,e^x - \int sin(x)e^x\,dx = sin(x)\,e^x - (-cos(x)e^x + \int cos(x)\,e^x\,dx\,)$$

$$= sin(x)\,e^x + cos(x)e^x - \int cos(x)\,e^x\,dx$$

It looks like we're not getting anywhere — but remember, integrals can be manipulated algebraically. For instance, review the derivation of the integration by parts formula

We could rewrite

$$\int cos(x)e^x\,dx = sin(x)\,e^x + cos(x)e^x - \int cos(x)\,e^x\,dx$$

As...

$2\int cos(x)e^x\,dx = sin(x)\,e^x + cos(x)e^x$, by adding $\int cos(x)e^x\,dx$ on both sides of the equation

Thus, it follows that...

$$\int cos(x)e^x\,dx = \frac{1}{2}\,sin(x)\,e^x + \frac{1}{2}\,cos(x)e^x = \frac{1}{2}\,e^x\,(sin(x) + cos(x)) + C$$

3. $x\,(ln\,(x) - 1\,) + C$ (final answer)

Steps:

u = $ln\,(x)$ v = x

du = $\frac{1}{x}$ dv = 1

$$\int ln(x)\,dx = x\,ln(x) - \int x\,(\tfrac{1}{x})\,dx = x\,ln(x) - x + C$$

INTEGRATION BY PARTIAL FRACTIONS:

By now, you should know that it is possible to express a fraction as a sum of two seperate, albeit not necessarily distinct, fractions, if the denominator can be factored.

For example, we can rewrite:
$\frac{3x+11}{x^2-x-6} = \frac{3x+11}{(x-3)(x+2)} = \frac{A}{(x-3)} + \frac{B}{(x+2)}$

Multiplying both sides of the denominator gives us
$3x + 11 = A\,(x+2) + B(x-3)$

And solving for A and B would yield that A = 4 and B = -1.* So,
$\frac{3x+11}{x^2-x-6} = \frac{4}{(x-3)} - \frac{1}{(x+2)}$

Why is this helpful? Well, let;s say we were asked to integrate f(x) = $\frac{3x+11}{x^2-x-6}$. Instinctively, we might try to compute this integral by using u-substitution, but unfortunately, we will not be able to do so in this case.
$\int \frac{3x+11}{x^2-x-6}\,dx$
Let's let $u = x^2 - x - 6$. Consequently, $du = 2x - 1\,dx$. Now what?

In order for us to use u-substitution, we would need to be able to be able to substitute $\frac{du}{2x-1}$ for dx, without altering the original function. Given that the numerator is a different polynomial function, we cannot easily perform such a substitution. So what can we do instead?

We have shown that $\frac{3x+11}{x^2-x-6} = \frac{4}{(x-3)} - \frac{1}{(x+2)}$. So, why not just **rewrite the integrand as the sum of two fractions**, as we know that $\int \frac{1}{x+c}\,dx = ln\,|x+c| + C_1$ (by the Chain Rule, differentiating $ln\,|x+c|$ is still equal to $\frac{1}{x+c}$, since the derivative of x+c is still just 1)

Thus,
$\int \frac{3x+11}{x^2-x-6}\,dx = \int \frac{4}{(x-3)} - \frac{1}{(x+2)}\,dx = 4\,ln\,|x-3| - ln\,|x+2| + C$. Simple enough, right?

If the denominator of the integrand is in the form f(x) = $\frac{P(x)}{Q(x)}$, and $Q(x)$ is factorable, we could write the fraction as the sum of two other fractions in order to simplify the integral. Note however, this might always be necessary --- for instance, $\int \frac{3x^2+3}{x^3+3x-6}\,dx$ (Can you tell why?)

This is essentially what **Integration By Partial Fractions** is -- reducing the integrals into a sum of two other integrals.

* We urge that you review partial fraction decomposition before continuing to solve any problems -- otherwise, you will likely have no clue on how to solve integrals by partial fraction decomposition, because obvious reasons.

Examples:
Try these problems by yourself first before reading the solutions below:

1. $\int \frac{7x}{x^2+14x+40}\,dx$
2. $\int \frac{4x+10}{(x-2)(x-1)^2}\,dx$
3. $\int \frac{4}{x^2-5x+14}\,dx$
4. $\int \frac{8+x+6x^2-12x^3}{(3x^2+4)\,(x^2+7)}\,dx$ **Hint:** $\int \frac{1}{u^2+1} = u'\ tan^{-1}(u) + C$
5. $\int \frac{3x^3+7x^2+x+9}{x+1}\,dx$ **Hint:** *long division*

Solutions:

1. $\frac{1}{3}(-14ln\,|x+4| + 35ln\,|x+10|) + C$

We can factor the denominator $x^2 + 14x + 40$ as $(x+4)\,(x+10)$, making the integral essentially

$\int \frac{7x}{(x+4)\,(x+10)}dx$

By partial fraction decomposition, we can rewrite $\int \frac{7x}{(x+4)\,(x+10)}dx$ as $\int \frac{A}{(x+4)} + \frac{B}{(x+10)}dx$

Multiplying both sides of the denominator gives us

$7x = A\,(x+10) + B(x+4)$

A = -28/6 = -14/3

B = 70/6 = 35/3

Thus, $\int \frac{7x}{(x+4)\,(x+10)}dx = \int \frac{-14}{3(x+4)} - \frac{35}{3(x+10)}dx = \frac{1}{3}(-14ln\,|x+4| + 35ln\,|x+10|) + C$

We can of course differentiate this to check:

$\frac{d}{dx}\frac{1}{3}(-14ln\,|x+4| + 35ln\,|x+10|) = \frac{1}{3}\left(\frac{-14}{(x+4)} + \frac{35}{(x+10)}\right) = \frac{1}{3}\left(\frac{-14x-140+35x+140)}{(x+4)(x+10)}\right) = \frac{1}{3}\left(\frac{21x}{(x+4)(x+10)}\right) = \frac{7x}{x^2+14x+40}$

2. $18\,ln\,|x-2| - 18\,ln\,|x-1| + \frac{14}{x-1} + C$

Rewrite the integral as follows:

$\int \frac{4x+10}{(x-2)(x-1)^2} dx = \int \frac{A}{(x-2)} + \frac{B}{(x-1)} + \frac{C}{(x-1)^2}$

$A\,(x-1)2 + B\,(x-2)\,(x-1) + C\,(x-2) = A\,(x-1)^2 + B\,(x-2)\,(x-1) + C\,(x-2)$

x = 1	$14 = -C \rightarrow C = -14$
x = 2	$18 = A$
x = 0	$10 = 18 + 2B + 28 \rightarrow B = -18$

$\int \frac{4x+10}{(x-2)(x-1)^2} dx = \int \frac{18}{(x-2)} - \frac{18}{(x-1)} - \frac{14}{(x-1)^2} dx = 18\,ln\,|x-2| - 18\,ln\,|x-1| + \frac{14}{x-1} + C$

3. $\frac{4}{9}(\,ln\,|\,x-7| - ln\,|x+2|\,) + C$

$\int \frac{4}{x^2-5x-14} dx = \int \frac{4}{(x-7)(x+2)} dx = \int \frac{4}{9\,(x-7)} - \frac{4}{9\,(x+2)} dx = \frac{4}{9}(\,ln\,|\,x-7| - ln\,|x+2|\,) + C$

4. $\frac{1}{2}\,ln\,|3x^2+4| - \frac{5}{2} ln\,|x^2+7| + \frac{2\sqrt{7}}{7} tan^{-1}\,(\frac{x}{\sqrt{7}}) + C$

5. $x^3 + 2x^2 - 3x + 12\,ln\,|x+1| + C$

This integral actually doesn't involve partial fraction decomposition --- but you can also simplify integrals by long division, if the degree of the numerator is greater than the degree in the denominator.

INDETERMINATE FORMS -- L'HÔPITAL'S RULE REVISITED

L'Hôpital's Rule tells us, if $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$, and f(x) and g(x) are differentiable at an open interval *I,* except at perhaps a in *I*, then $\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$, provided that $\lim_{x\to a} \frac{f'(x)}{g'(x)}$ exists (and is not an indeterminate form).

Confused? Go back a few sections and revisit L'Hôpital's Rule; from here on out, I will assume that you are familiar with L'Hôpital's Rule.

Sidenote: For the sake of typing convenience, I will, from now on, write "L'Hôpital" as "L'Hospital" --the mathematician in question reportedly spelled his name that way, and I'm also too tired to insert an "ô" on my American keyboard every 2 seconds.

L'Hospital's Rule works great if the limit yields $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$. But unfortunately, there are quite a few more indeterminate forms you may encounter in single-variable calculus. These are:

$0(\infty)$, $\infty - \infty$, 1^∞, 0^0 and ∞^0

So, how do we evaluate a limit that yields one of the above indeterminate forms? We can still use L'Hospital's Rule -- all we have to do is rewrite the function as a quotient, so that we can see that the limit is equal to $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$ (so that we can use L'Hospital's Rule).

Too wordy/confusing? <u>Let's take a look at a few examples:</u>

1. $\lim_{x\to -\infty} xe^x$

Evaluating this limit gives us $\lim_{x\to -\infty} xe^x = 0(-\infty)$, which is slightly problematic.

We could rewrite $\lim_{x\to -\infty} xe^x$ as $\lim_{x\to -\infty} \frac{e^x}{\frac{1}{x}}$,which is equivalent --- just a slightly more complicated way of denoting the limit.

Then, $\lim_{x\to -\infty} \frac{e^x}{\frac{1}{x}} = \frac{0}{0}$, which we can evaluate using L'Hopital's Rule!

$\lim_{x\to -\infty} \frac{e^x}{\frac{1}{x}} = \lim_{x\to -\infty} \frac{e^x}{\frac{-1}{x^2}}$ which is... still indeterminate. Which means we have to continue evaluating the limit!

$\lim_{x\to -\infty} \frac{e^x}{\frac{-1}{x^2}} = \lim_{x\to -\infty} \frac{e^x}{\frac{2}{x^3}} = \lim_{x\to -\infty} \frac{e^x}{\frac{-6}{x^4}} = \lim_{x\to -\infty} \frac{e^x}{\frac{24}{x^5}}$ ... still indeterminate.

Let's try switching the numerator and denominator to see if that helps, since in this form, we will always still get an indeterminate form.

$\lim_{x\to -\infty} xe^x = \lim_{x\to -\infty} \frac{x}{\frac{1}{e^x}} = \frac{0}{0}$.

Now, by L'Hospital's Rule, $\lim_{x\to -\infty} \frac{1}{\frac{-1}{e^x}} = 0$.

And we're done! $\lim_{x\to -\infty} xe^x = 0$

Case 1: $0(\infty)$

When you're evaluating a limit in the form $\lim_{x\to a} f(x)\,g(x)$ such that the limit yields $0\,(\infty)$, re-express the limit as $\lim_{x\to a} \frac{f(x)}{\frac{1}{g(x)}}$ or $\lim_{x\to a} \frac{g(x)}{\frac{1}{f(x)}}$, depending on which form is more simplistic/yields a determinate value.

2. $\lim_{x\to 1+} \frac{x}{x-1} - \frac{1}{ln(x)}$

Evaluating this limit gives us $\lim_{x\to 1+} \frac{x}{x-1} - \frac{1}{ln(x)} = \infty - \infty$.

To write this as a quotient, let's try to find a **common denominator** such that we only have one expression to deal with.

$\lim_{x\to 1+} \frac{x}{x-1} - \frac{1}{ln(x)} = \frac{xln(x) - x + 1}{(x-1)ln(x)}$

Now, evaluating the limit gives us $\lim_{x\to 1+} \frac{xln(x) - x + 1}{(x-1)ln(x)} = \frac{0}{0}$.

By L'Hopital's Rule, $\lim_{x\to 1+} \frac{xln(x) - x + 1}{(x-1)ln(x)} \overset{\text{l'hop}}{=} \lim_{x\to 1+} \frac{1-ln(x) - 1}{1+ln(x) - \frac{1}{x}} = \lim_{x\to 1+} \frac{ln(x)}{1+ln(x) - \frac{1}{x}} \overset{\text{l'hop}}{=} \lim_{x\to 1+} \frac{\frac{1}{x}}{\frac{1}{x} + \frac{1}{x^2}} = ½$.

So, $\lim_{x\to 1+} \frac{x}{x-1} - \frac{1}{ln(x)} = \frac{3}{2}$

3. $\lim_{x\to+\infty} (x^2 + 3x + 7)^{0.5} - x$

If one of the terms is a square root, multiply by the conjugate:

$\lim_{x\to+\infty} \sqrt{x^2 + 3x + 7} - x = \lim_{x\to+\infty} \frac{\sqrt{x^2+3x+7} + x\,(\sqrt{x^2+3x+7} - x)}{\sqrt{x^2+3x+7} + x} = \lim_{x\to+\infty} \frac{3x+7}{\sqrt{x^2+3x+7} + x} = \frac{\pm\infty}{\pm\infty}$

$\lim_{x\to+\infty} \frac{3x+7}{\sqrt{x^2+3x+7} + x} \overset{\text{l'hop}}{=} \lim_{x\to+\infty} \frac{3}{\frac{2x+3}{2\sqrt{x^2+3x+7}} + 1} = \frac{3}{2}$

Case 2: $\infty - \infty$

When you're evaluating a limit in the form $\lim_{x\to a} f(x) - g(x)$ such that the limit yields $\infty - \infty$, either express the limit as one term or multiply the expression by the conjugate, in the event of a square root function.

4. $\lim_{x\to+\infty} (1 + \frac{4}{x})^x$

This limit is of the form $\mathbf{1^\infty}$. In this scenario, it's better to define the function **in terms of another variable**, then take the natural logarithm of both sides.

$y = (1 + \frac{4}{x})^x$

$\lim_{x\to+\infty} ln\,(y) = \lim_{x\to+\infty} x\,ln\,(1 + \frac{4}{x})$, since $log\,(a^b) = b\,log\,(a)$

Now, we can express the right-hand side in terms of a quotient: $x\,ln\,(1 + \frac{4}{x}) = \frac{ln(1+\frac{4}{x})}{\frac{1}{x}}$

$$\lim_{x\to+\infty} \frac{ln(1+\frac{4}{x})}{\frac{1}{x}} = \frac{0}{0}$$

$$\lim_{x\to+\infty} \frac{ln(1+\frac{4}{x})}{\frac{1}{x}} = \text{l'hop} \lim_{x\to+\infty} \frac{\frac{-4}{(1+4/x)x^2}}{\frac{-1}{x^2}} = \lim_{x\to+\infty} \frac{4}{(1+\frac{4}{x})} = 4.$$

So, it looks like we have finished evaluating this limit --- but remember that this whole time, we weren't solving for $\lim_{x\to+\infty} (1+\frac{4}{x})^x$, but rather $\lim_{x\to+\infty} x\, ln\,(1+\frac{4}{x})$, or $\lim_{x\to+\infty} ln\,(y)$.

Since $y = (1+\frac{4}{x})^x$, we need to solve for y.

Conveniently enough, we know that $ln\,(y) = 4$. So, $y = e^4$

Our final answer is $\lim_{x\to+\infty} (1+\frac{4}{x})^x = e^4$

Case 3, 4, 5: 1^∞, 0^0, and ∞^0

When you're evaluating a limit in the form $\lim_{x\to a} f(x)^{g(x)}$ such that the limit yields 1^∞, 0^0, or ∞^0, set the limit equal to y, take the natural logarithm of both sides, and solve *that* limit. Then solve for y separately, once you've evaluated the limit.

Remember, not everything is an indeterminate form! If you apply L'Hopital's Rule to a limit that is not of an indeterminate form, the resultant limit will be different from the original limit you were trying to solve for!

Determinate-Indeterminate Forms Table	
Indeterminate Forms	Determinate Forms
$0/0$	$\infty+\infty=\infty$
$\pm\infty/\pm\infty$	$-\infty-\infty=-\infty$
$\infty-\infty$	$0^\infty=0$
$0(\infty)$	$0^{-\infty}=\infty$
0^0	$(\infty)\cdot(\infty)=\infty$
1^∞	
∞^0	
Use L'Hôpital's Rule	Do *Not* Use L'Hôpital's Rule

ARC LENGTH

In Algebra I, you should have learned how to calculate the distance between 2 points (x_1, y_1), (x_2, y_1) on a Cartesian plane.

Just as a quick refresher, the distance between any two points (x_1, y_1) and (x_2, y_1) can be determined as follows:

Distance Formula:

Just as a quick refresher, the distance between any two points (x_1, y_1) and (x_2, y_1) can be determined as follows:

$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$, derived from the Pythagorean Theorem.

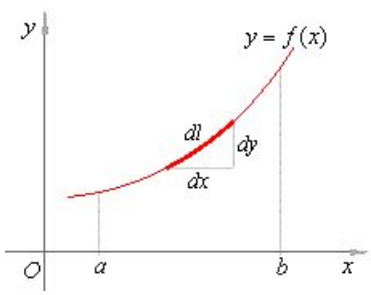
To calculate the length of a given arc, we use the formula below:

The length of any given arc bounded by *a* and *b* can be written as follows:

$$\textbf{\textit{arc length}} = \int_a^b \sqrt{1 + (\tfrac{dy}{dx})^2}\, dx$$

Read this if you are interested in knowing why the formula works, if not you can skip to the examples in which this formula is applied.

To see why this is the case, what we're doing is essentially summing an infinite number of straight lines that form the curve, almost analogous to summing the area of an infinite number of rectangles to calculate the area under a curve.



Let's establish a set of given points on the curve bounded below and above by *a* and *b*, respectively. Given 2 arbitrary points P_i and P_{i-1}, the distance between the points can be expressed as follows:

$|P_i - P_{i-1}|$

If we write the curve as the sum of an infinite straight lines, the length of the curve can be expressed as:

$$\lim_{n\to\infty} \sum_{i=1}^{n} \left|(P_i - P_{i-1}\right|$$

Now, expressing P_i, P_{i-1} in terms of the coordinates (x, y), and knowing the distance formula, we can rewrite the above formula as:

$$\lim_{n\to\infty} \sum_{i=1}^{n} \left|\sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2}\right|$$

By MVT, there exists a point $x \in [x_{i-1}, x_i]$ such that $f'(x) = \frac{y_i - y_{i-1}}{x_i - x_{i-1}}$, or

$f'(x)(x_i - x_{i-1}) = y_i - y_{i-1}$.

So,

$$\lim_{n\to\infty} \sum_{i=1}^{n} \left| \sqrt{(x_i - x_{i-1})^2 + (\tfrac{dy}{dx})^2 (x_i - x_{i-1})^2} \right|$$

$$\lim_{n\to\infty} \sum_{i=1}^{n} \left| (x_i - x_{i-1}) \sqrt{1 + (\tfrac{dy}{dx})^2} \right|$$

Taking the limit as n approaches infinity (remember that dx and $\frac{dy}{dx}$ represent the changes in x and y respectively) yields the following:

$$\int_a^b \sqrt{1 + (\tfrac{dy}{dx})^2}\, dx$$

1. Find the length of the arc of $f(x) = x^{3/2}$ from x = 1 to x = 8
2. Find the length of the arc of $f(x) = 2y^{3/2}$ from y = 0 to y = 1

Solutions:

1. $\int_1^8 \sqrt{1 + (\frac{3}{2}x^{1/2})^2}\, dx = \int_1^8 \sqrt{1 + (\frac{9}{4}x)}\, dx = \frac{8}{27}(1 + \frac{9}{4}x)^{3/2} \Big|_1^8 = 22.803$
2. $\int_0^1 \sqrt{1 + (3y^{1/2})^2}\, dy = \int_0^1 \sqrt{1 + 9y}\, dy = \frac{2}{27}(1 + 9y)^{3/2} \Big|_0^1 = 2.27$

IMPROPER INTEGRALS:

There are situations in which, when computing a definite integral, one or both of the limits of integration are infinity and/or there exists an infinite discontinuity. In such a scenario, we can simply substitute an arbitrary variable for the bound in question, and take the limit as that variable approaches infinity.

In other words, if we were asked to find the integral of *f* bounded from *a* to +∞, assuming that a is a constant and *f* is continuous from [a, +∞):

$$\int_a^{\infty} f(x)\,dx = \lim_{b\to\infty^-} \int_a^b f(x)\,dx = \lim_{b\to\infty^-} [F(x)]\,\Big|_a^b = \lim_{b\to\infty^-} [F(x) - F(a)]$$

If b is constant, and we instead were asked to find the integral of *f* bounded from -∞ to *b*, under the assumption that *f* is continuous from [a, +∞):

$$\int_{-\infty}^{b} f(x)\,dx = \lim_{a\to -\infty^+} \int_a^b f(x)\,dx = \lim_{a\to -\infty^+} [F(x)]\,\Big|_a^b = F(b) - \lim_{a\to -\infty^+} F(a)$$

Note that the limit might not exist, which would imply that the integral does not exist, or *diverges (*does not have a finite value).

For instance, let $f(x) = \frac{1}{x}$, and let's integrate from 1 to positive infinity:

$$\int_1^{\infty} \frac{1}{x}\,dx = \lim_{b\to\infty^-} \int_1^b f(x)\,dx = \lim_{b\to\infty^-} ln(x)\Big|_1^b = \lim_{b\to\infty^-} ln(b) - ln(1) = \lim_{b\to\infty^-} ln(b)$$

Great! But what's $\lim_{b\to\infty^-} ln(b)$?

It is simply ∞.

So our solution to this integral is ... ∞?

Remember, infinity is not a *number* - it is a concept. We can't have the definite integral of a function be infinity, only a definable constant value. Think about it visually: how would you say that the area under a curve is infinite?

We can't so therefore, we say that this integral ***diverges.***

If *a* and *b* are -∞ and ∞ respectively, we must determine a point *c*, where *c* is a constant within the interval (-∞, +∞), and **integrate the 2 functions separately**. Like this:

$$\int_{-\infty}^{\infty} f(x)\,dx = \lim_{a\to -\infty^+} \int_a^c f(x)\,dx + \lim_{b\to\infty^-} \int_c^b f(x)\,dx = \lim_{a\to -\infty^+} [F(x)]\,\Big|_a^c + \lim_{b\to\infty^-} [F(x)]\,\Big|_c^b$$

If either $\lim_{a\to -\infty^+} [F(x)]\,\Big|_a^c$ OR $\lim_{b\to\infty^-} [F(x)]\,\Big|_c^b$ *diverges,* then $\int_{-\infty}^{\infty} f(x)\,dx$ likewise *diverges* as well.

However, just like when a limit of integration is infinity, the integral may not *necessarily exist* if there is an infinite discontinuity in the interval in which we are integrating ; that is, it *diverges*. If you look at the graph of

$f(x) = \frac{1}{x}$, there exists a vertical asymptote at x = 0. Let's say we were to integrate $\frac{1}{x}$ from some negative number *a*, and some positive number *b:*

$$\int_a^b \frac{1}{x} dx = \lim_{c \to 0^-} \int_a^c \frac{1}{x}\, dx + \lim_{c \to 0^+} \int_c^b \frac{1}{x}\, dx = \lim_{c \to 0^-} [ln\,|x|]|_a^c + \lim_{c \to 0^+} [ln\,|x|]|_c^b\ ) =$$

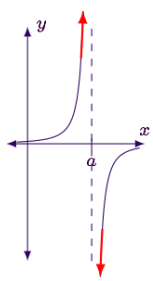
$$(\lim_{c \to 0^-} [ln\,|c| - ln\,|a|]) + (\lim_{c \to 0^-} [ln\,|b| - ln\,|c|]).$$

Slight problem: What is $\lim_{c \to 0^-} ln\,|c|$?

If you take a look at the graph of $\frac{1}{x}$, we see that $\lim_{c \to 0^-} ln\,|c| = -\infty$.

Likewise, $\lim_{c \to 0^-} ln\,|c| = +\infty$

And since we can't have a definite integral be equal to ∞, we can say that this integral ***diverges.***



If there exists an infinite discontinuity at x = *c* on the interval [a, b], where a and b are constants, and *f* is otherwise continuous on [a,b], then we must evaluate the integral as follows:

$$\int_a^b f(x)\, dx = \lim_{x \to c^-} \int_a^c f(x)\, dx + \lim_{x \to c^-} \int_c^b f(x)\, dx = \lim_{x \to c^-} [F(x)]\,|_a^c + \lim_{x \to c^-} [F(x)]\,|_c^b$$

For example, in the graph on the right, if we were to integrate the function from *x = 0* to some *x = b*, such that *b* is a positive constant greater than *a*, the integral of that function (let's call it *g*, and the antiderivative *G*) would be written as follows:

$$\int_0^b f(x)\, dx = \lim_{x \to a^-} \int_0^a f(x)\, dx + \lim_{x \to a^-} \int_a^b f(x)\, dx = \lim_{x \to a^-} [F(x)]\,|_0^a + \lim_{x \to a^-} [F(x)]\,|_a^b$$

There are 4 required steps for improper integrals:

1. **Rewrite** improper integral as a limit
2. **Antiderivative** with limits
3. **FTOC 1** - evaluate limits
4. **Check** if the limit actually makes sense y'all - like, obviously you can't have the final answer be infinity something. Like, $\frac{1}{\infty}$ Is fine, but infinity in the final answer? No.

If the limit equals a finite value, then the integral converges to that value. If the limit does not equal a finite value, then the integral diverges

Sidenote:

What is the derivative of e^x ?

$$\lim_{h\to 0} \frac{e^{x+h} - e^x}{h} = \lim_{h\to 0} \frac{e^x(e^h - 1)}{h} = e^x \lim_{h\to 0} \frac{(e^h - 1)}{h}$$

Let $y = e^h - 1$

$h = ln(y+1)$

$\lim_{h\to 0} e^h - 1 = 0$

Substitute:

$$e^x \lim_{y\to 0} \frac{(e^{ln(y+1)} - 1)}{ln\,(y+1)} = e^x \lim_{y\to 0} \frac{y}{ln\,(y+1)} = e^x \lim_{y\to 0} \frac{1}{ln\,((y+1)^{\frac{1}{y}})} = e^x \frac{1}{ln(e)} = e^x$$

What is the derivative of $ln\,(x)$?

$$\lim_{h\to 0} \frac{ln(x+h) - ln(x)}{h} = \lim_{h\to 0} ln\,(\frac{x+h}{x})^{1/h} = \lim_{h\to 0} ln\,(1 + \frac{h}{x})^{1/h}$$

Let $y = \frac{h}{x}$, $\lim_{h\to 0} \frac{h}{x} = 0$

$$\lim_{y\to 0} ln\,(1 + y)^{1/yx} = \frac{1}{x} \lim_{y\to 0} ln\,(1 + y)^{1/y} = \frac{1}{x} .$$

Note that the derivative of $ln\,(x)$ is restricted to the interval $(0, \infty)$.

Yet the derivative of $ln\,(-x)$ is still $\frac{1}{x}$, using the same substitution as above:

$$\lim_{h\to 0} \frac{ln(-x-h) - ln(-x)}{h} = \lim_{h\to 0} ln\,(1 + \frac{h}{x})^{1/h} = \frac{1}{x} .$$

What is the antiderivative of $\frac{1}{x}$?

For $x \in (0, \infty)$, $\int \frac{1}{x} dx = ln(x) + C, \ x > 0$

For $x \in (-\infty, 0)$, $\int \frac{1}{x} dx = ln(-x) + C, \ x < 0$

Combining the two answers, we can say that $\int \frac{1}{x} dx = ln|x| + C$